# Data Flow Diagram (DFD)

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| **Date** | 18 October 2026 |
| **Team ID** | Rahul-DS633 |
| **Project Name** | CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction |
| **Maximum Marks** | 5 Marks |

## Level 0: Context Data Flow & External Interfaces

| Entity | Data Inflow to System | Data Outflow from System |
|---|---|---|
| **Investigator** | • Crime Scene Photos (JPG/PNG)<br>• 911 Audio / Witness Tapes (WAV/MP3)<br>• CCTV Surveillance Video (MP4)<br>• Natural Language Questions | • Interactive 3D Spatial Crime Scene<br>• Second-by-Second Synced Timeline<br>• Grounded AI Answers & Summaries<br>• Court-Admissible PDF Dossiers |
| **Model Registry** | • Model weights (RT-DETR, CLIP, Whisper)<br>• Historical case sessions JSON | • Session context & timeline records<br>• Generated PDF reports in /data/reports/ |
| **Groq Cloud API** | • LLaMA-3.3 forensic hypotheses<br>• Contextual answers & 4 translations | • Structured prompt payloads containing detected objects and timeline keypoints |

## Level 1: Functional Subsystem Data Flow Decomposition

| Process | Input | Algorithm / Engine | Output Destination |
|---|---|---|---|
| **P1: Vision** | 2D Scene Image | RT-DETR object detection + Depth estimation [X,Y,Z]. | BBoxes, 3D Mesh -> SessionStore |
| **P2: Audio** | Audio Waveform | Whisper ASR Speech-to-Text + Groq translation. | Transcript, translations -> SessionStore |
| **P3: Video** | CCTV Video | 1.0 Hz Frame sampler + BLIP captioning + Anomaly. | 1Hz Timeline -> SessionStore |
| **P4: Fusion** | P1, P2, P3 data | Cross-modal timestamp synchronization & consistency. | Unified Dossier -> SessionStore |
| **P5: PDF** | Session JSON | ReportLab Platypus synthesis + XML character sanitation. | PDF File -> /data/reports/ |